1. Introduction

1.1 Purpose

The purpose of this Data Dictionary is to define the entities, attributes, relationships, primary keys, foreign keys, and data types used within the Big Brother Student Accommodation Management System database. The database supports both the Flutter mobile application and the ASP.NET web application through a shared Supabase backend.

1.2 Scope

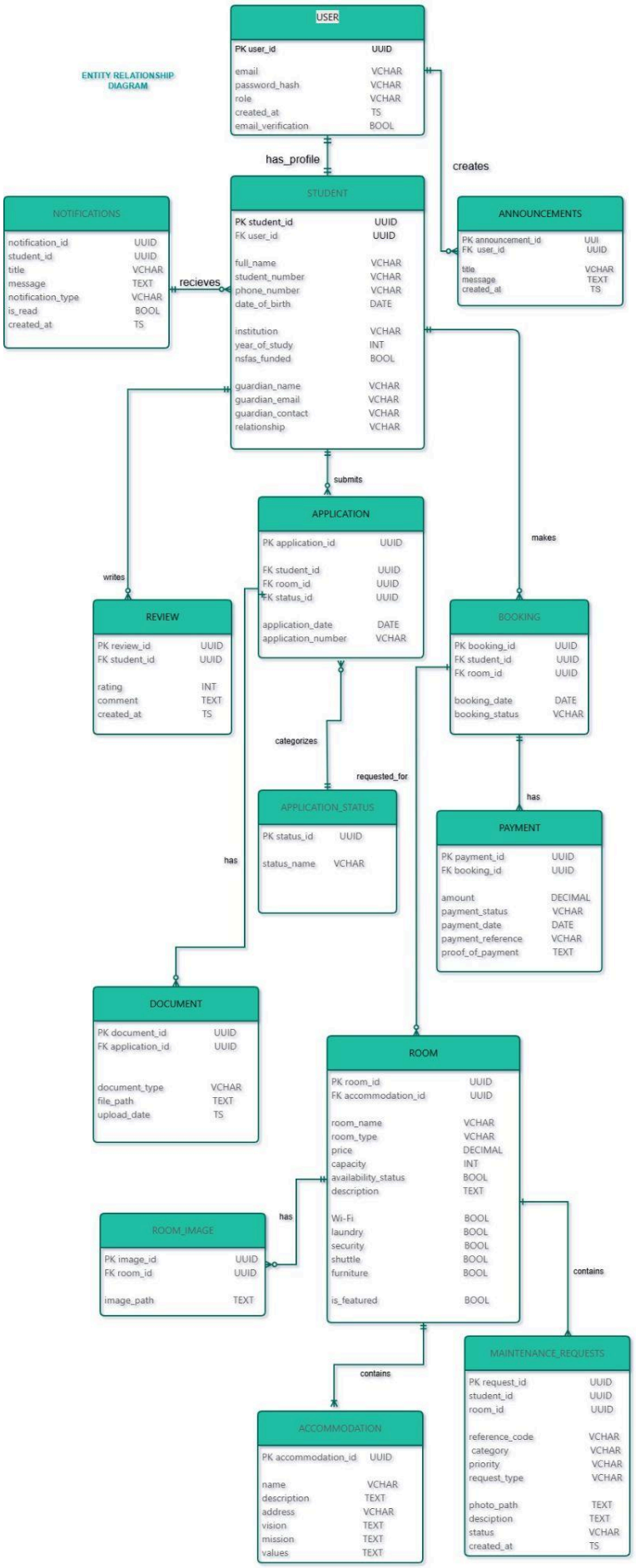
This document covers all database components required to support:

- User authentication
- Student registration
- Accommodation management
- Room management
- Applications
- Bookings
- Document uploads
- Payments
- Reviews
- Maintenance requests
- Announcements
- Notifications

These database structures are derived from the approved requirements, use cases, sequence diagrams, and UI wireframes.

Data Type	Description
UUID	Unique identifier
VARCHAR(n)	Variable length text
TEXT	Large text field
INTEGER	Whole number
BOOLEAN	True or False value
DECIMAL (10,2)	Numeric value with decimals
DATE	Calendar date
TIMESTAMP	Date and time value

Entity Relationship Diagram



3. Entity Definitions

3.1 USER

Stores authentication and account information for all system users.

Attribute	Type	Key	Description
user_id	UUID	PK	Unique user identifier
email	VARCHAR (255)		User email address
password_hash	VARCHAR (255)		Encrypted password
role	VARCHAR (20)		User role
created_at	TIMESTAMP		Account creation date

USER (1:1) STUDENT

- One USER account belongs to one STUDENT.

3.2 STUDENT

Stores personal, academic, and guardian information for students applying for accommodation.

Attribute	Data Type	Key	Description
student_id	UUID	PK	Unique identifier for each student
user_id	UUID	FK	References USER
full_name	VARCHAR(100)		Student full name
student_number	VARCHAR (20)		University student number
phone_number	VARCHAR (20)		Student contact number
date_of_birth	DATE		Student date of birth
institution	VARCHAR (100)		Current institution attended
year_of_study	INTEGER		Academic year of study

nsfas_funded	BOOLEAN		Indicates NSFAS funding status
guardian_name	VARCHAR(100)		Parent or guardian name
guardian_email	VARCHAR(255)		Parent or guardian email
guardian_contact	VARCHAR(20)		Parent or guardian contact numbe
relationship	VARCHAR(50)		Relationship to student

USER (1:1) STUDENT

STUDENT (1:M) APPLICATION

STUDENT (1:M) BOOKING

STUDENT (1:M) REVIEW

STUDENT (1:M) MAINTENANCE_REQUEST

- Each STUDENT is linked to one USER account.
- One STUDENT can submit multiple APPLICATIONS.
- One STUDENT can make multiple BOOKINGS.
- One STUDENT can submit multiple REVIEWS.
- One STUDENT can submit multiple MAINTENANCE_REQUESTS.
- Each MAINTENANCE_REQUEST belongs to one STUDENT.

3.3 ACCOMMODATION

Stores information about the accommodation facility.

Attribute	Data Type	Key	Description
accommodation_id	UUID	PK	Unique accommodation identifier
name	VARCHAR(100)		Accommodation name
description	TEXT		Accommodation description
address	VARCHAR(255)		Physical address
vision	TEXT		Organization vision statement
mission	TEXT		Organization mission statement
values	TEXT		Organization values

ACCOMMODATION (1:M) ROOM

- One ACCOMMODATION contains multiple ROOMS.
- Each ROOM belongs to one ACCOMMODATION.

3.4 ROOM

Stores room information and accommodation facilities.

Attribute	Data Type	Key	Description
room_id	UUID	PK	Unique room identifier
accommodation_id	UUID	FK	References ACCOMMODATION
room_name	VARCHAR(50)		Room name or number
Room_type	VARCHAR(50)		Type of room
price	DECIMAL(10,2)		Monthly room price
capacity	INTEGER		Maximum number of occupants
availability_status	BOOLEAN		Indicates room availability
description	TEXT		Room description
wifi	BOOLEAN		Indicates Wi-Fi availability
laundry	BOOLEAN		Indicates laundry availability
security	BOOLEAN		Indicates security availability
shuttle	BOOLEAN		Indicates shuttle availability
furniture	BOOLEAN		Indicates furniture availability

ACCOMMODATION (1:M) ROOM

ROOM (1:M) APPLICATION

ROOM (1:M) BOOKING

ROOM (1:M) MAINTENANCE_REQUEST

ROOM (1:M) ROOM_IMAGE

- Each ROOM belongs to one ACCOMMODATION.
- One ROOM may appear in multiple APPLICATIONS.
- One ROOM may have multiple BOOKINGS over time.

• One ROOM can have multiple MAINTENANCE_REQUESTS over time. • Each MAINTENANCE_REQUEST is associated with one ROOM. • One ROOM can have multiple ROOM_IMAGES.

• Each ROOM_IMAGE belongs to one ROOM.

3.5 APPLICATION

Stores accommodation applications submitted by students.

Attribute	Data Type	Key	Description
application_id	UUID	PK	Unique application identifier
student_id	UUID	FK	References STUDENT
room_id	UUID	FK	References ROOM
status_id	UUID	FK	References APPLICATION_STATUS
application_no	VARCHAR(50)		Public application reference number
application_date	DATE		Date application was submitted

STUDENT (1:M) APPLICATION

ROOM (1:M) APPLICATION

APPLICATION_STATUS (1:M) APPLICATION

APPLICATION (1:M) DOCUMENT

- One STUDENT can submit multiple APPLICATIONS.
- One ROOM may be associated with multiple APPLICATIONS.
- Each APPLICATION has one APPLICATION_STATUS.
- One APPLICATION may contain multiple DOCUMENTS.

3.6 APPLICATION_STATUS

Stores application status values used in tracking applications.

Attribute	Data Type	Key	Description
status_id	UUID	PK	Unique status identifier
status_name	VARCHAR(20)		Status description

APPLICATION_STATUS (1:M) APPLICATION

- One APPLICATION_STATUS may be linked to many APPLICATIONS.
- Each APPLICATION has one status.

3.7 DOCUMENT

Stores documents uploaded during the application process.

Attribute	Data Type	Key	Description
document_id	UUID	PK	Unique document identifier
application_id	UUID	FK	References APPLICATION
document_type	VARCHAR(50)		Type of document uploaded
file_path	TEXT		Path to stored document
upload_date	TIMESTAMP		Date and time document was uploaded

APPLICATION (1:M) DOCUMENT

- One APPLICATION may contain multiple DOCUMENTS.
- Each DOCUMENT belongs to one APPLICATION.

3.8 BOOKING

Stores room booking information for successful accommodation applicants.

Attribute	Data Type	Key	Description
booking_id	UUID	PK	Unique booking identifier
student_id	UUID	FK	References STUDENT
room_id	UUID	FK	References ROOM
booking_date	DATE		Date booking was created
booking_status	VARCHAR(20)		Current booking status

STUDENT (1:M) BOOKING

ROOM (1:M) BOOKING

BOOKING (1:M) PAYMENT

- One STUDENT may have multiple BOOKINGS.
- One ROOM may have multiple BOOKINGS.

• One BOOKING may have multiple PAYMENT records.

3.9 PAYMENT

Stores payment records associated with accommodation bookings.

Attribute	Type	Key	Description
payment_id	UUID	PK	Unique payment identifier
booking_id	UUID	FK	References BOOKING
payment_reference	VARCHAR(30)		Unique payment reference number
payment_method	VARCHAR(50)		Payment method selected
proof_of_payment	TEXT		Uploaded proof of payment file
reported_amount	DECIMAL(10,2)		Amount reported by student
reported_date	DATE		Date payment was reported
verification_status	VARCHAR(20)		Verification status of payment
due_date	DATE		Payment due date
payment_date	DATE		Actual payment date

BOOKING (1:M) PAYMENT

- Each PAYMENT belongs to one BOOKING.
- One BOOKING may be associated with multiple PAYMENTS.

3.10 REVIEW

Stores student ratings and feedback regarding accommodation services. Supported by the Review & Feedback screen.

Attribute	Data Type	Key	Description
review_id	UUID	PK	Unique review identifier
student_id	UUID	FK	References STUDENT
rating	INTEGER		Rating score
comment	TEXT		Student feedback
created_at	TIMESTAMP		Date review was submitted

STUDENT (1:M) REVIEW

- One STUDENT may submit multiple REVIEWS.
- Each REVIEW belongs to one STUDENT.

3.11 MAINTENANCE_REQUEST

Attribute	Type	Key	Description
request_id	UUID	PK	Unique maintenance request identifier
student_id	UUID	FK	References STUDENT
room_id	UUID	FK	References ROOM
reference_code	VARCHAR(30)		Maintenance request reference number
category	VARCHAR(50)		Category of maintenance issue
priority	VARCHAR(20)		Priority level of issue
request_type	VARCHAR(30)		Type of maintenance request
description	TEXT		Description of maintenance issue
photo_path	TEXT		Uploaded maintenance image
status	VARCHAR (20)		Current request status
created_at	TIMESTAMP		Date and time request was created

STUDENT (1:M) MAINTENANCE_REQUEST

ROOM (1:M) MAINTENANCE_REQUEST

Each MAINTENANCE_REQUEST belongs to one STUDENT.

One STUDENT can submit multiple MAINTENANCE_REQUESTS.

Each MAINTENANCE_REQUEST is associated with one ROOM.

One ROOM can have multiple MAINTENANCE_REQUESTS over time.

3.12 NOTIFICATION

Attribute	Type	Key	Description
notification_id	UUID	PK	Unique notification identifier
student_id	UUID	FK	References STUDENT
title	VARCHAR (100)		Notification title
message	TEXT		Notification content
notification_type	VARCHAR (50)		Type of notification
is_read	BOOLEAN		Indicates whether notification was read
created_at	TIMESTAMP		Date and time notification was created

STUDENT (1:M) NOTIFICATION

- Each NOTIFICATION belongs to one STUDENT.
- One STUDENT can receive multiple NOTIFICATIONS.

3.13 ANNOUNCEMENT

Attribute	Type	Key	Description
announcement_id	UUID	PK	Unique announcement identifier
user_id	UUID	FK	References USER
title			

- One USER may create multiple ANNOUNCEMENTS.
- Each ANNOUNCEMENT is created by one USER.

3.14 ROOM_IMAGE

Attribute	Type	Key	Description
image_id	UUID	PK	Unique image identifier
room_id	UUID	FK	References ROOM
image_path	TEXT		Storage location of room image

- One ROOM can have multiple ROOM_IMAGES.
- Each ROOM_IMAGE belongs to one ROOM.

4. Relationship Summary

Parent Entity	Cardinality	Child Entity	Description
USER	1:1	STUDENT	Each student has one user account, and each user account belongs to one student.
USER	1:M	ANNOUNCEMENT	management user can create multiple announcements.
STUDENT	1:M	APPLICATION	A student can submit multiple accommodation applications.
APPLICATION_STATUS	1:M	APPLICATION	One status can be associated with multiple applications.

ROOM	1:M	APPLICATION	One room can appear in multiple applications over time.
APPLICATION	1:M	DOCUMENT	An application can contain multiple supporting documents.
STUDENT	1:M	BOOKING	A student can make multiple bookings.
ROOM	1:M	BOOKING	A room can have multiple bookings over time.
BOOKING	1:M	PAYMENT	A booking can have multiple payment records.
STUDENT	1:M	REVIEW	A student can submit multiple reviews and feedback entries

ACCOMMODATION	1:M	ROOM	One accommodation property contains multiple rooms.
ROOM	1:M	ROOM_IMAGE	A room can have multiple images associated with it.

STUDENT	1:M	MAINTENANCE_REQUEST	A student can submit multiple maintenance requests.
ROOM	1:M	MAINTENANCE_REQEST	A room can have multiple maintenance requests overtime.
STUDENT	1:M	NOTIFICATION	A student can receive multiple notifications.

Total Entities: 14

5. Database Normalization

The database design follows normalization principles to reduce data redundancy and improve data consistency. Separate entities have been created for applications, bookings, documents, notifications, maintenance requests, and room images to ensure efficient data management and maintain referential integrity.

6. Database Rules / Constraints

1. A student must have a registered user account before submitting an application.
2. A room must belong to an accommodation.
3. An application must be linked to one student and one room.
4. A booking can only exist for a valid room.
5. A payment must be linked to an existing booking.
6. A maintenance request must be linked to both a student and a room.
7. Notifications are linked to individual students.
8. Room images cannot exist without a room.